Quiz for Clean Kerala:A Statewide Water Quality & Sanitation Challenge

For Higher Secondary Students (10th + 1, +2 of Kerala , CBSE, ICSE) Syllabus

Preamble

Water quality and sanitation are among the most pressing concerns of our time. With the increasing incidence of water contamination and related issues, especially in regions like Kerala where traditional water sources such as wells and borewells are widely used, there is a growing need for awareness and education on these topics.

This programme is designed with the **objective of creating awareness among the student community about the importance of water quality and sanitation**. Recognizing the role of youth in driving sustainable change, the initiative seeks to equip students with the knowledge and tools necessary to understand and address water-related challenges in their surroundings. We are also prepared to provide study materials and facilitate interactive sessions for students who are interested in taking up assignments or projects related to water quality and sanitation. This awareness programme is a collaborative effort jointly organized by the **Centre for Water Research and Education (CWRE), Unnat Bharath Abhiyan (UBA), and the Indian Society for Technical Education (ISTE).**

The CWRE, functioning under the **Department of Civil Engineering at Jyothi Engineering College**, is actively engaged in research, education, and outreach activities focused on water and sanitation. Our core mission includes **Information, Education, and Communication (IEC)** in the water sector, aimed at empowering communities to address water-related challenges. We operate a government-approved commercial water testing laboratory and offer water testing services at a nominal fee. In addition, CWRE conducts training programmes, publishes books and articles, and undertakes various community-oriented initiatives related to water conservation, quality monitoring, and sanitation awareness. We hope that this initiative will inspire students to become proactive contributors in promoting safe water practices and sustainable sanitation solutions in their communities.

For details visit our website cwre.jeccac.in

Modality and Eligibility for the Water Quality & Sanitation Awareness Examination

- **Eligibility Criteria:**
  - Students currently studying in 10th standard ,Plus One (+1) and Plus Two (+2) under the Kerala State, CBSE, or ICSE Syllabus are eligible to participate.
  - Only students from Kerala state are permitted to appear for the examination.
  - Participants must be currently enrolled in school and must produce a valid identity proof at the time of the examination.
- **Examination Details:**
  - The exam will be conducted in offline in OMR sheet at Jyothi Engineering College. Students are required to make their own travel arrangements.
  - The question paper will consist of 100 multiple choice questions (MCQs).
    - Out of these, certain questions will be starred and used as tie-breakers if needed.
    - 100 sample questions are provided in advance, and 60 of these will appear in the final exam.
  - The exam key will be published within 5 hours of completion of the examination.
  - A formal challenge window will be provided for students to contest the answer key, within a specified time frame.
- **Registration and Fee:**
  - An **entrance fee of ₹100** will be charged per candidate.
  - Amount may be paid and details entered using the QR code given below

- **Awards and Prizes:**
  - Attractive cash prizes will be awarded to the top performers:
    - **First Prize: ₹10,000**
    - **Second Prize: ₹7,000**
    - **Third Prize: ₹5,000**

- All prize winners will also receive Certificates of Merit online
- Those who score 50% or above will receive Consolation Certificate online

- **Additional Information:**
  - Students may contact the Centre for Water Research and Education (CWRE) for any clarification or assistance.
  - In case of any disputes, the decision of the Administrative Committee will be final.
  - Further details will be made available through the official CWRE website
  - Tentative date for the written exam is 17th October 2026

Office bearers for the programme are as follows:

Sl.No	Name and Designation	Phone number	
1	Fr. David Nettikkaden , Executive Manager		
2	Dr. P Sojanlal, Principal		
3	Dr. Alwyn Varghese, HOD CE		
4	Prof. Cyriac.M.G, General Convener	9447732619	Cyriacmg@jecc.ac.in
5	Ms. Anna Joseph, Convener	9400947041	annajoseph@jecc.ac..in
6	Mr. Suhas Nair S, Convener	6381153948	ssuhasnair@jecc.ac.in
7	Ms. Jeffy Johny, Convener	7025168862	jeffyjohny@jecc.ac.in

(NB: The faculties mentioned in4,5,6,7,8 may be contacted for any clarifications)

Syllabus

Topic: Water Quality and Sanitation Awareness

This syllabus is designed to provide students with a fundamental understanding of water quality, sanitation, and related public health and environmental concerns. The content is aimed at creating awareness and promoting informed action among young learners.

1. Water Quality and Sanitation: An Overview

- Global scenario of water quality and sanitation
- Relevance of water quality monitoring in daily life and health

2. Water Quality measurement and Parameters

- Physical Parameters:
  - pH
  - Turbidity
  - Conductivity
  - Total Dissolved Solids (TDS)
- Chemical Parameters

    - Chloride
    - Alkalinity
    - Hardness
    - Nitrate
    - Sulphate
    - Chemical Oxygen Demand (COD)
    - Biochemical Oxygen Demand (BOD)
- Biological Parameters:
    - Coliform bacteria
    - E. coli contamination

3. Contamination and Pollution

- Heavy metal contamination in water
- Pesticide pollution and its effects
- Sources and effects of contamination

4. Waterborne and Water-based Diseases

- Common diseases:
    - Cholera
    - Typhoid
    - Leptospirosis
    - Other bacterial and viral infections

5. Water Quality Issues in Kerala

- Local challenges in water quality
- Community-based and technological remedies

6. Water Treatment Methods

- Filtration
- Disinfection and Chlorination
- Use of water treatment kits and basic household practices
- Commercial water treatment systems

7. Sanitation and Waste Management

- Household Sanitation Systems:
    - Septic tank
    - Leach pit
    - Soak pit
- Solid Waste Management at Household Level
- Role of Local Self Government Departments (LSGD) in managing solid and liquid waste
- Personal and environmental hygiene
- Maintenance of home and surroundings
- Importance of regular hand washing

8. National and Global Initiatives

- Swachh Bharat Abhiyan (Clean India Mission) and unnat bharath abhiyan
- Sustainable Development Goals (SDGs) on water and sanitation

9. Fundamentals of air pollution

10. National and global level initiates for pollution control and functioning of national green tribunal

11. Air pollution problems , Global warming and Remedial measures

12. Gifts of the nature like hydrogeological cycle ,lapse rate bacterial life etc that benefit the human beings

13. Major sanitation issues in Kerala state

This syllabus form the basis for the multiple-choice questions (MCQs) in the examination. Students are encouraged to explore each topic for both theoretical understanding and practical relevance.

For any clarification or study support, students may contact the **Centre for Water Research and Education (CWRE).**

Hundred 100 model questions are given below and out of that at least 60 will appear in the final exam

Jyothi Engineering College (AUTONOMOUS)
Accredited by NAAC (A Grade) and
Programmes Accredited by NBA (CE, CS, EC, EE, ME, MR)
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Approved by AICTE & Affiliated to APJ Abdul Kalam Technological University
A Centre of Excellence in Science and Technology by the Catholic Archdiocese of Trichur

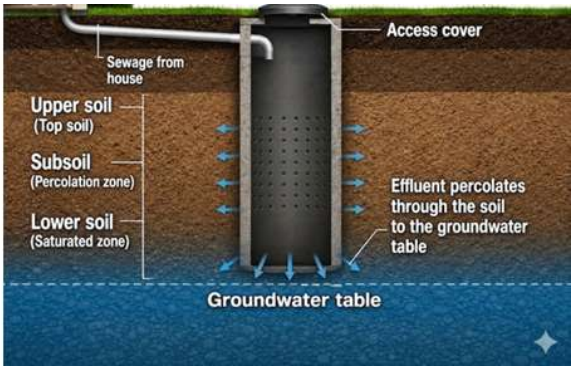
Water Quality Quiz for 10th +1,+2 students_2026-27

Exercise for students: There are 100 questions. Out of this 60 will appear in the final exam.

1. Malaria is spread by which microorganism?
2. Leptospirosis is caused by ……
3. Why we consider below given living being as a dreadful one?



4. What is the difference between typhoid and typhus?
5. What is the symptom for west Nile fever?
6. What is the most dangerous form of hepatitis?
7. Where do we find that growth of gappi fish is meritorious?
8. What is the basic difference between Coliform and Ecoli?
9. Popular discovery of Dr.Ronald Ross is……..
10. How does Dengu fever spread?
11. What is the significance of permissible and acceptable limit in water quality?
12. What do the below picture indicate?

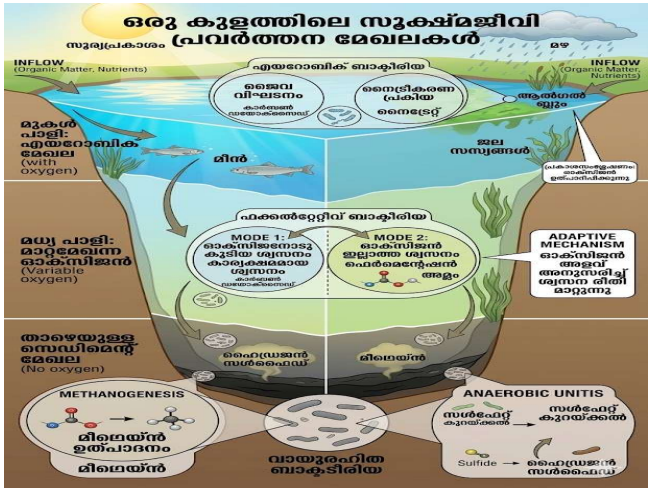


13. The more I study nature, the more I stand amazed at the work of the creator. who said these words and whom he referred as the creator?
14. What way chlorine tablets differ from bleaching powder?

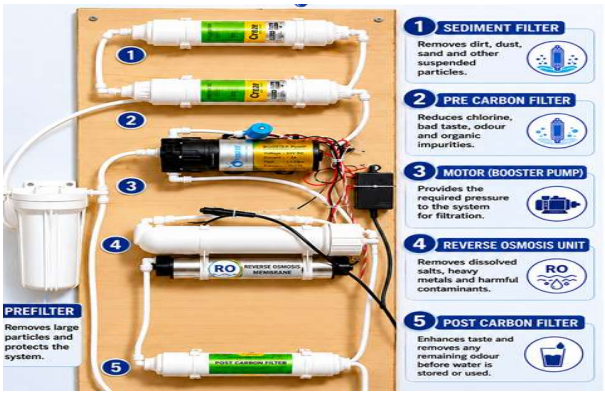
15. Which is the organization responsible for prescribing water quality standards in India?
16. NEERI is located in which city of India.
17. Which ministry of GOI is responsible to deal with environmental protection in India?
18. Why do we prefer water tanks of the type given below?



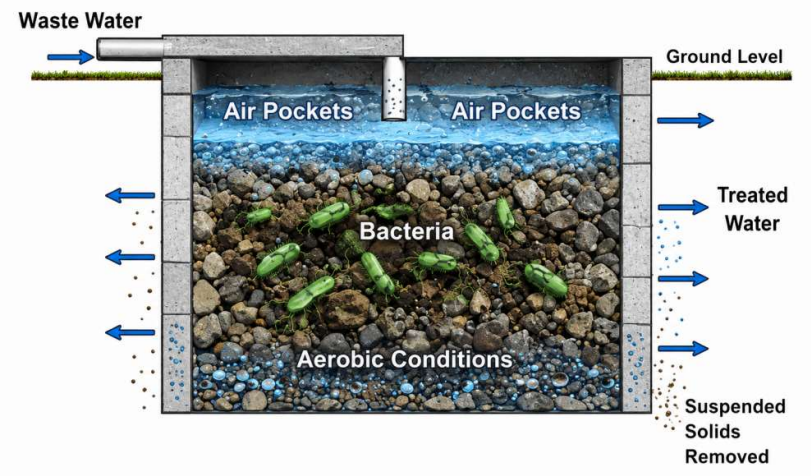
19. What is the process given below?



20. In the bottom there is oxygen is limited and production of methane gas takes place. Point out the popular system working under the above principle which is regularly used for housed purposes
21. What is the commercial name of the instrument given below and its use?



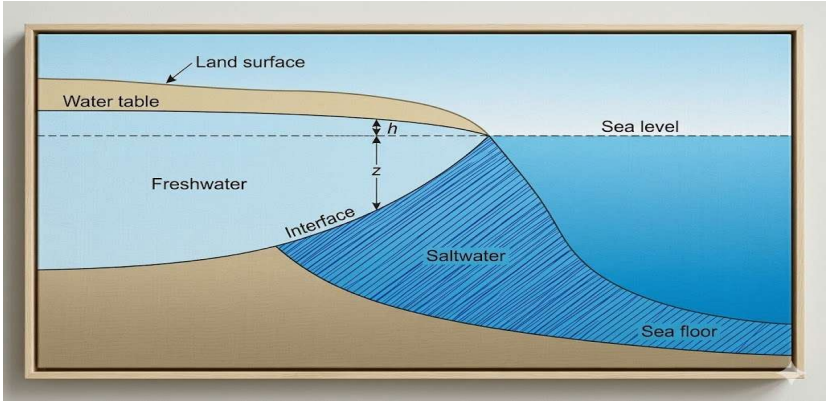
22. What is the principle behind waste water treatment in the unit given below?



23. What is the basic difference between soak pit and soak trench?
24. What is the major disadvantage of the building given below in respect of air purity?



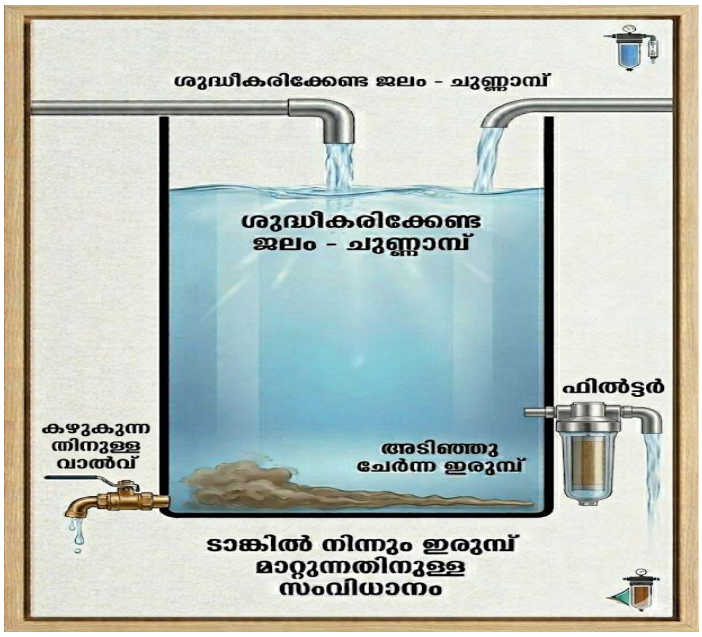
25. Sea water intrusion is a major issue in coastal area. What does the below picture indicate?



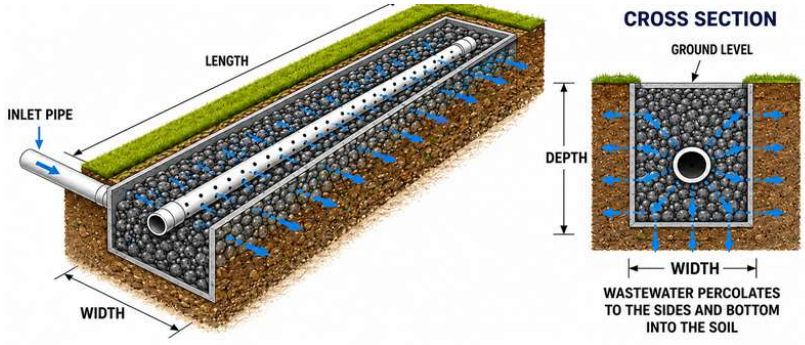
26. What is a suitable measure for controlling sea water intrusion?
27. Why do we say that a water tank should always be accessible a house?
28. What is Robert Koch is popular for?
29. What is the main problem with plastic pollution?
30. What does the below figure indicate?



31. What does the picture below indicate?



32. Where doses RO plants are mainly used?
33. What does "biodegradable" mean?
34. What does the below picture indicate
35. What is the function of the system given below and its name?



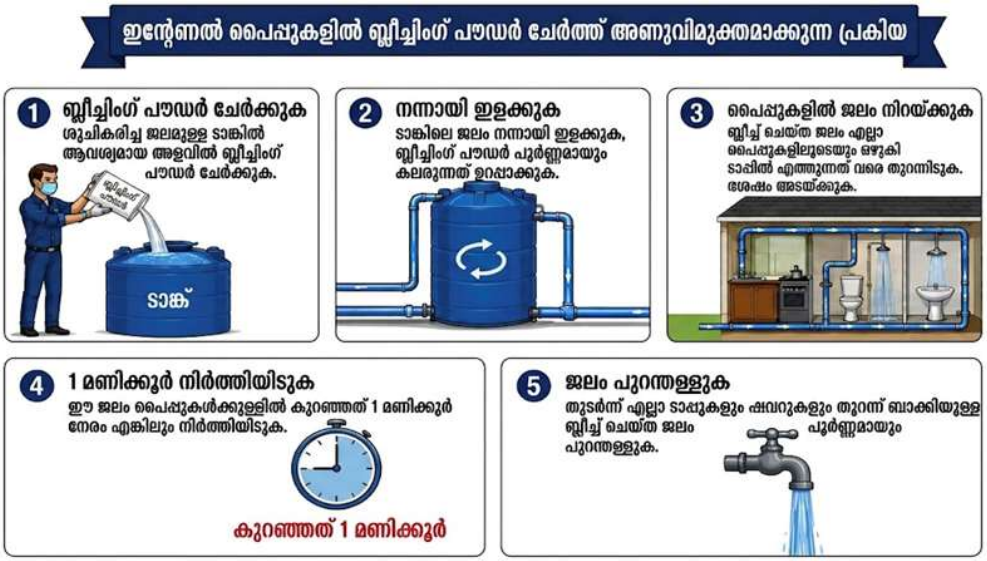
36. What are the main sources of hazardous waste in household ?
37. Organically contaminated well water contains more iron. Why ?
38. ICMS stands for…….. and what is its use ?
39. Shredder is a term connected with……
40. Which SDG Is connected with health , water and sanitation?
41. National Green Tribunal (NGT) Act was implemented in which year.
42. Bacteria Shigella causes …….. Disease.
43. Who is the inspiration behind the concept of SBM?
44. Which ministry initiated the Swachch Bharat Mission in India?

45. World hand wash day is celebrated on.....
46. In winter air pollution effects are more why?
47. There is a very elaborative system to ensure pollution free air to the world .Name the process.
48. What is meant PM 10 in air pollution.
49. What does ammonia in drinking water indicates?
50. Which method is commonly used to remove Fluoride from drinking at domestic level?
51. What is the by product from the system given below and how it is beneficial to us?

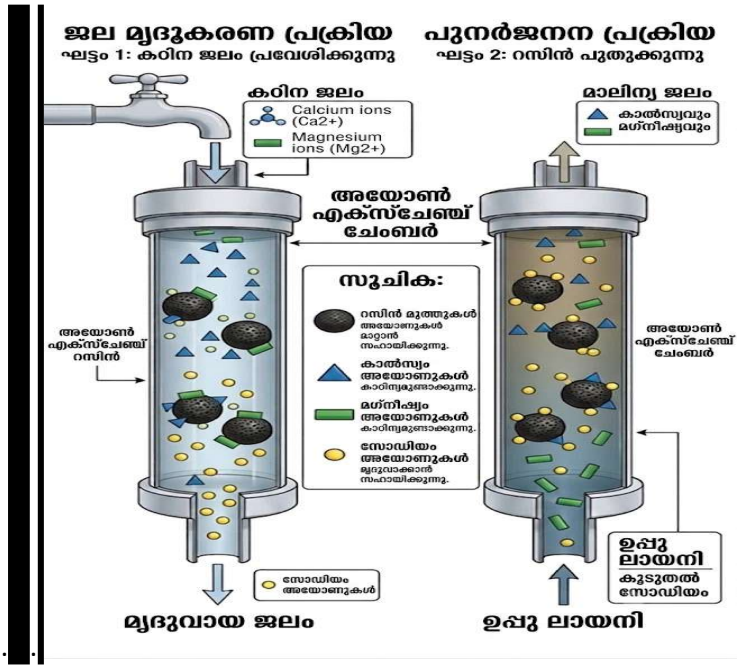


52. What is the relation between hardness and calcium?
53. Residual chlorine of 1.5 indicates.....
54. What does NABL stands for?
55. Dmitry Ivanosky is popular in ..... Discipline.
56. Good portion of air pollutants are related to oxygen. Why ?

Carefully study the picture below and answer the questions



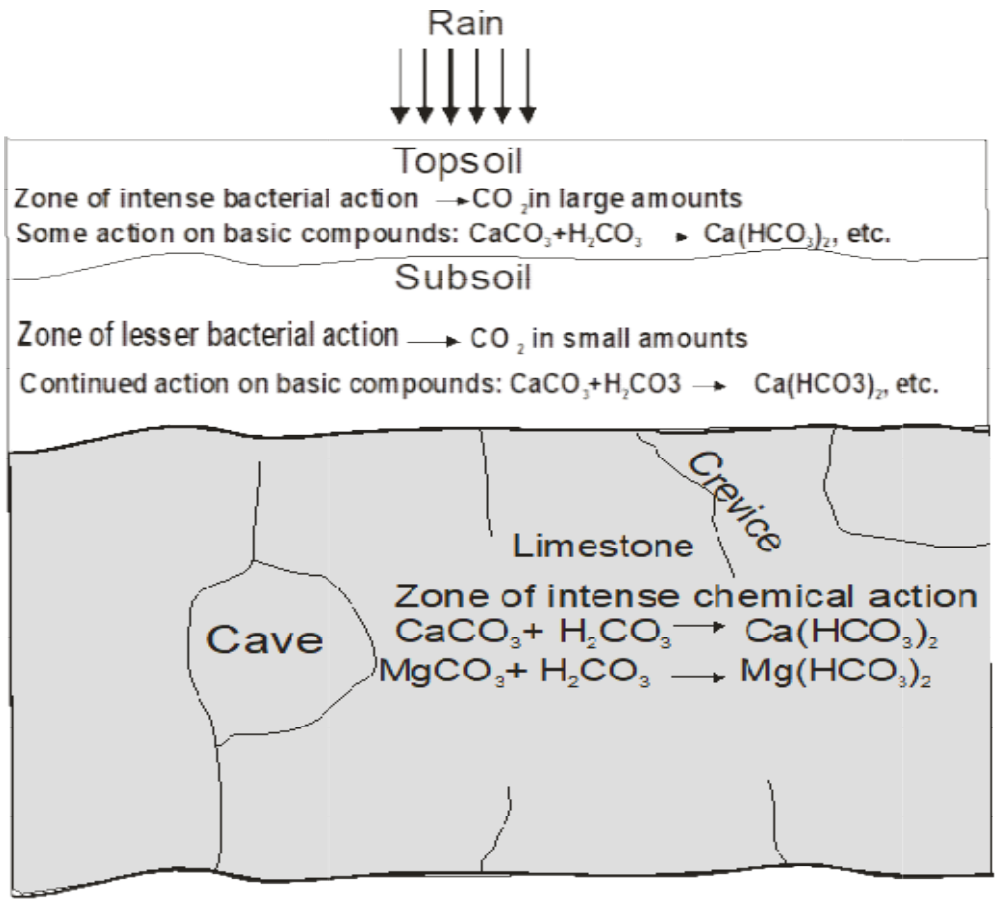
57. Under which condition these treatments are appropriate
58. What is the main advantage of the above treatment system?



59. What is the purpose of the instrument given above?
60. What is the role of sodium ion in this process?

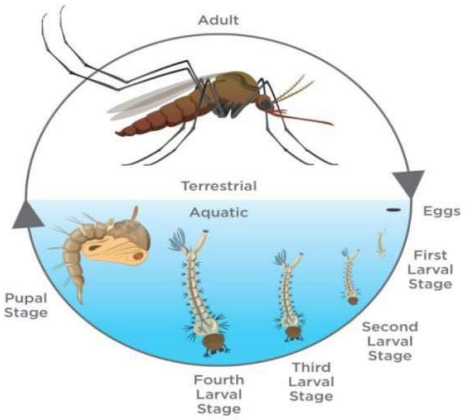
61. Why regeneration is carried out using salt water?

Below picture shows the chemical process while rain water is percolated through the soil. Carefully study the picture and answer the following questions from



62. List any a probable water quality issue in a shallow well.
63. What are the alkaline substances present in a deep water sources?
64. Why magnesium carbonate is less in borewell water compared to calcium carbonate?
65. If pH is very high in a shallow well ,What do you assess?
66. In a water sample pH , hardness iron are high. What do you normally interpret?
67. Louis Paster is popular for what. Which country he belongs to?
68. What is the permissible water quality standard for chloride?

69. What does the below picture indicate?



70. Change in colur of water sample upon boiling indicates.
71. What are the benefit of hand washing with soap?
72. Black color for water well water indicate.
73. What is Colorimetry?
74. Why do we say that trees are beneficial to mankind
75. What does wash stand for in the context of water , sanitation and hygiene
76. What is the reason for blue baby syndrome
77. White deposit on boiling in water indicates presence of.....
78. Presence of organic matter in water is established by the analysis of
79. Sour taste in water indicates
80. What is the purpose of Life straw?
81. pH of a solution is 2 at normal tempurature. What is the pOH ?
82. What is grey water?
83. Which microorganisms are commonly used as an indicator of faecal contamination in water?
84. What is the role of wetlands in maintaining water quality?
85. Why is source protection important for groundwater quality?
86. Why biodegradable matter is more relevant for human life?
87. Indian leader who took a prominent role for 1972 Stockholm conference.
88. Who discovered that cholera is transmitted through contaminated water.
89. Which scientist is known as the “father of modern microbiology”?
90. What is eutrophication of a water body?
91. What is the basic difference between disinfection by chlorination and ozone
92. What is the principle mechanism for functioning of a septic tank?
93. Theory of ionization was proposed by.....

94. List two methods other than chlorination suitable for controlling algae growth in wells.
95. What is the general application of Composting?
96. What is the natural underground storage of water called?
97. Why is disinfection necessary after filtration of drinking water?
98. What is the importance of residual Chlorine after disinfection?
99. How does very low ph is identified.
100. What is the basic difference between disinfection by Bleaching powder and Chlorine gas?